SENTHIL PUBLIC SCHOOL, DHARMAPURI

AFFILIATED TO CBSE, NEW DELHI • UNIFIED PHYSICAL BINDER & TOPPER BENCHMARK

STUDENT: Tharun	**CLASS: Class 8 CBSE**	**SUBJECT: Mathematics**
EXAM ID: EXAM-V3-VERIFY-001	**DATE: 18-10-2026**	**SCORE: 31.5 / 35.0 (90.0%) - Grade A1**

PART 1: EVALUATED SCORECARD & RED-INK CORRECTED ANSWER SCRIPT

☐☐ Active Learning Verdict: VERIFIED: Tharun demonstrated high cognitive retention of all study slabs. Conceptual clarity is strong.

Q.1 to Q.5: Section A: Objective & Assertion-Reasoning [5 Marks] **5.0 / 5.0**

Tharun's Script: 1.(b) $(a-b)^2=a^2-2ab+b^2$; 2.(c) Pascal; 3.(c) Drip Irrigation; 4.(c) Permanent Settlement; 5.(a) Both A and R are true and R explains A.

Evaluator Red-Ink Annotations: [+5.0] Perfect concept identification in Section A.

Q.6: Factorise by grouping: 15xy - 6x + 5y - 2 [2 Marks] **2.0 / 2.0**

Tharun's Script: = 3x(5y - 2) + 1(5y - 2) = (5y - 2)(3x + 1)

Evaluator Red-Ink Annotations: [+2.0] Clean factorization by grouping.

Q.9: Calculate maximum pressure exerted by wooden block 20x10x5 cm of mass 4 kg [3 Marks] **2.5 / 3.0**

Tharun's Script: F = m*g = 4*10 = 40 N. Min Area = 10cm * 5cm = 0.10 * 0.05 = 0.005 m^2. P = F/A = 40 / 0.005 = 8000.

Evaluator Red-Ink Annotations: [+2.5] Excellent formula application and conversion to meters.
[-0.5 UNIT MISSING] Wrote 8000 without 'Pascals' (Pa). Never omit SI units!

Q.10: Perimeter of pool is 154 m. Length is 2m more than 2b. Find l and b [3 Marks] **3.0 / 3.0**

Tharun's Script: Let b = x m. l = (2x + 2) m. P = 2(l + b) => 154 = 2(2x + 2 + x) => 154 = 2(3x + 2) => 77 = 3x + 2 => 3x = 75 => x = 25 m. Breadth = 25 m, Length = 2(25)+2 = 52 m.

Evaluator Red-Ink Annotations: [+3.0] Flawless step-by-step algebraic proof. Both values boxed with units.

Q.12: Cylindrical storage tank radius 7m, height 3m. Sheet required & cost [5 Marks] **4.5 / 5.0**

Tharun's Script: Total Surface Area = 2*pi*r*(r + h) = 2 * (22/7) * 7 * (7 + 3) = 44 * 10 = 440 m^2. Cost = 440 * 45 = ☐ 19,800.

Evaluator Red-Ink Annotations: [+4.5] Correct formula and arithmetic.
[-0.5] Missing concluding word statement: 'Total sheet required is 440 m^2 and total expenditure is ₹ 19,800'.

PART 2: SIDE-BY-SIDE 'HOW THARUN ANSWERED' VS. 'CBSE TOPPER MODEL ANSWER'

Comparative diagnostic identifying exact presentation gaps, missing keywords, and formula layout.

Q.1 to Q.5: Section A: Objective & Assertion-Reasoning [5 Marks]

Tharun's Approach	CBSE Topper Benchmark (Full Marks)
Tharun's Written Answer: 1.(b) $(a-b)^2=a^2-2ab+b^2$; 2.(c) Pascal; 3.(c) Drip Irrigation; 4.(c) Permanent Settlement; 5.(a) Both A and R are true and R explains A. Slip/Gap: None. Flawless recall.	CBSE Topper Model Answer (100% Marks): Option letter followed by full text correctly stated. Keywords to Underline: Identity $(a-b)^2$, Pascal, Drip Irrigation, Permanent Settlement

Q.6: Factorise by grouping: 15xy - 6x + 5y - 2 [2 Marks]

Tharun's Approach	CBSE Topper Benchmark (Full Marks)
Tharun's Written Answer: = 3x(5y - 2) + 1(5y - 2) = (5y - 2)(3x + 1) Slip/Gap: Good layout. Always write '=' at start of each line.	CBSE Topper Model Answer (100% Marks): 15xy - 6x + 5y - 2 = 3x(5y - 2) + 1(5y - 2) = (5y - 2)(3x + 1). Boxed final answer. Keywords to Underline: Common factor (5y - 2)

Q.9: Calculate maximum pressure exerted by wooden block 20x10x5 cm of mass 4 kg [3 Marks]

Tharun's Approach	CBSE Topper Benchmark (Full Marks)
Tharun's Written Answer: F = m*g = 4*10 = 40 N. Min Area = 10cm * 5cm = 0.10 * 0.05 = 0.005 m^2. P = F/A = 40 / 0.005 = 8000. Slip/Gap: Omitted S.I. unit next to 8000. CBSE spot-evaluators deduct 0.5 mark immediately.	CBSE Topper Model Answer (100% Marks): Maximum Pressure (P) = Force / Minimum Area = 40 N / 0.005 m^2 = 8000 Pa (or N/m^2). Boxed: [P_{max} = 8,000 Pa]. Keywords to Underline: P = F / A, Minimum Surface Area, 8,000 Pa

Q.10: Perimeter of pool is 154 m. Length is 2m more than 2b. Find l and b [3 Marks]

Tharun's Approach	CBSE Topper Benchmark (Full Marks)
Tharun's Written Answer: Let b = x m. l = (2x + 2) m. P = 2(l + b) => 154 = 2(2x + 2 + x) => 154 = 2(3x + 2) => 77 = 3x + 2 => 3x = 75 => x = 25 m. Breadth = 25 m, Length = 2(25)+2 = 52 m. Slip/Gap: None. Excellent work.	CBSE Topper Model Answer (100% Marks): Breadth = 25 m, Length = 52 m. Verification: 2(52 + 25) = 2(77) = 154 m. Keywords to Underline: Let breadth = x m, Perimeter = 2(l + b), 25 m, 52 m

Q.12: Cylindrical storage tank radius 7m, height 3m. Sheet required & cost [5 Marks]

Tharun's Approach	CBSE Topper Benchmark (Full Marks)
Tharun's Written Answer: Total Surface Area = 2*pi*r*(r + h) = 2 * (22/7) * 7 * (7 + 3) = 44 * 10 = 440 m^2. Cost = 440 * 45 = ☐ 19,800. Slip/Gap: Calculation was correct but missed the formal concluding statement required in 5M CBSE questions.	CBSE Topper Model Answer (100% Marks): 1. TSA = $2\pi r(r + h)$ = 440 m^2. 2. Total Cost = Area x Rate = 440 x 45 = ☐ 19,800. Statement: Hence, 440 m^2 of sheet is needed at total cost of ☐ 19,800. Keywords to Underline: Total Surface Area = $2\pi r(r+h)$, 440 m^2, ☐ 19,800

PART 3: 'HOW TO STRUCTURE THE ANSWER' (P-E-E-D PROTOCOL) & KEYWORDS CHECKLIST

CBSE Board Presentation Standards & Mandatory Terminology Verification

CBSE TOPPER 5-POINT PRESENTATION PROTOCOL:

1. P-E-E-D Layout: Point (clear statement) ➔ Explain (concept) ➔ Evidence/Formula (equations with SI units) ➔ Diagram (clean pencil sketch).
2. Formula Declaration: State formula in standard symbols before substituting numbers (e.g., P = F / A).
3. Boxed Final Answer: Draw pencil box around every final numerical answer with SI unit: [Area = 150 cm²].
4. Mandatory Underlining: Underline core scientific/mathematical terminology with pencil on first mention.
5. Rough Work Column: Maintain a neat 4 cm right margin for rough calculations. Never scribble over it.

MANDATORY KEYWORDS CHECKLIST (FOR THARUN'S PHYSICAL REVISION BINDER)

Q.Num	Core Topic / Concept	Mandatory Keywords to Underline	Mastered
Q.1 to Q.5	Section A: Objective & Assertion-Reasoning [5...	Identity $(a-b)^2$, Pascal, Drip Irrigation, Permanent Settlement	[ ] Done
Q.6	Factorise by grouping: 15xy - 6x + 5y - 2 [2 ...	Common factor (5y - 2)	[ ] Done
Q.9	Calculate maximum pressure exerted by wooden ...	P = F / A, Minimum Surface Area, 8,000 Pa	[ ] Done
Q.10	Perimeter of pool is 154 m. Length is 2m more...	Let breadth = x m, Perimeter = 2(l + b), 25 m, 52 m	[ ] Done
Q.12	Cylindrical storage tank radius 7m, height 3m...	Total Surface Area = $2\pi r(r+h)$, 440 m^2, ☐ 19,800	[ ] Done